

TABLE IV
Hot Tweezers

Specifications	Manufacturer				
	OK International	HAKKO	Srinivasa Electronics	Metcal	Weller
Model	MFR-1140	Hakko FM-2022	PLUTO 555	MFR-1140	WMRT
Power consumption (W)	70	140	80	70	2×40
Temperature range (°C)	Smearheat-based variable power system	—	180-350	—	50-450
Tip-to-ground resistance (Ω)	< 2	< 2	< 2	< 2	—
Tip-to-ground potential (mV)	< 2	< 2	0.2	< 2	—
Heating element	Patented Smearheat based heater	Composite heater	Ceramic type	—	—
Cord length (m)	1.22	1.2	1.2	1.83	1 (approx.)
Dimensions (mm)	90×142×197	length - 148	—	90×142×197	—
Weight (gm)	120	64	2000	2300	—
ESD-safe (Yes/No)	Yes	Yes	Optional	Yes	Yes

suitable range. The wider the range, the better.

Temperature stability. This is a measure of how well the control circuit works, allowing for the heat losses into the environment and yet keeping the tip's temperature stable. The stability specification is valid only when the iron is at rest, not when it is actually being used for soldering.

Nozzle-to-ground resistance. Tip-to-ground potential is also called

Thermal recovery. This is also a measure of how well the control circuit works. When a joint is soldered, heat is lost from the tip through conduction and the tip's temperature drops. Thermal recovery refers to how long the soldering station takes with a particular tip at, say, room temperature, to get back to the initial temperature. The better the thermal recovery, the faster the operator can work.

The characteristics in Fig. 4 were obtained by the manufacturer (HAKKO) by melting a solder consecutively on a 5×5mm PCB land size and simultaneously measuring the tip's temperature with a thermocouple fitted on the tip.

ESD-safe. If you are interested in soldering a lot of static-sensitive parts such as CMOS chips or MOSFET transistors, you should look for slightly more advanced de-soldering stations that use static-dissipative material in their construction to ensure that static does not build up on the equipment itself. You may find these listed as 'ESD safe.' The cheap variants won't necessarily be ESD-safe but, nevertheless, will still probably perform perfectly well in most hobby or educational applications, if you take the usual anti-

static precautions when handling the components.

Hot-air rework system

To de-solder SMD components, hot-air rework systems are used. How

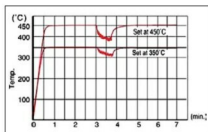


Fig. 4: Thermal recovery

millivolt leakage. This is the voltage existing between the soldering iron tip and the workstation common point ground. It should not exceed 2mV as defined by various standards.

Nozzle-to-ground potential. This is the resistance that exists between the tip of the soldering iron and the workstation common point ground. It should be between 2 ohms and 5 ohms.



Fig. 5: Hot-air rework system

they work is exactly opposite to how vacuum-pump type systems operate. Hot-air rework systems blow hot air on the target component, melting the solder. Once the solder has melted, the component is removed using regular tweezers or positioner. You will still have some solder on the pads, which is then removed by the vacuum pump type de-soldering system. Fig. 5 shows such a rework station.